UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

# Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 006 · Particular integral Type 4 ($e^{ax}V$) and Type 5 (xV) · compiled 16 October 2026

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| | |
|---|---|
| ESA ★★★ | Asked repeatedly in past ESA papers |
| ESA ★★ | Asked in past ESA papers or computed in class-work problems |
| ESA ★ | Conceptual: taught in class, not examined so far |
| ADDED FOR CLARITY | Standard context that is not in the class material |
| TRANSCRIPT ONLY | Said in the lecture but not on any slide |
| VIDEO ONLY | Seen only in a frame of the class video |
| NOT ON CURRENT SLIDES | From older material or past papers, not this year's slides |
| S006·5 | Unit 2 deck 006 (“Particular integral of Type 4”), slide 5 (rendered .pptx) |
| T006 00:04:12 | Unit 2 AV-summary transcript “Particular integral of standard functions – Type 4” (17 min), segment start |
| T006b 00:12:47 | Unit 2 AV-summary transcript “Particular integral – Type 3 and Type 4” (bucket 006, 15.5 min); only its Type 4 part (00:09:55–00:15:25) is cited here. The class 005 pack cites the same file's Type 3 part as T006a. |
| CW U2 P18 | Unit 2 class-work problem 18 |
| ESA Jan-24 s3 2c | Past paper: UE23MA141A Jan 2024 set 3, question 2.c |

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# A Formula sheet

## 1 The Type 4 rule: $X = e^{ax}V$ (exponential shift)

### E1.1 · Type 4: the exponential shift

ESA ★★

$$\frac{1}{f(D)}\big(e^{ax}V\big) = e^{ax}\,\frac{1}{f(D+a)}\,V = e^{ax}\big[f(D+a)\big]^{-1}V$$

GIVES The PI of $e^{ax}$ times any function $V$.

APPLIES $X = e^{ax}V$; $a$ = coefficient of $x$ in the exponent.

NOTE Write $e^{ax}$ to the left first, then replace every $D$ by $D+a$; $V$ is untouched.

S006·4 · S006·10 · T006 00:00:52 · T006 00:01:00 · T006b 00:10:19 · ESA Jan-24 s3 2c

### E1.2 · Why it works: $D$ past $e^{ax}$

ESA ★

$$D\big(e^{ax}V\big) = e^{ax}(D+a)V, \qquad f(D)\big(e^{ax}V\big) = e^{ax}f(D+a)V$$

GIVES The shift rule in the forward direction; inverting both sides gives E1.1.

APPLIES Product rule, repeated.

NOTE $D^2(e^{ax}V) = e^{ax}(D+a)^2V$, and so on for every power.

ADDED FOR CLARITY S006·4

### E1.3 · Then $V = \sin bx$ or $\cos bx$ (Type 2)

ESA ★★

$$e^{ax}\,\frac{1}{f(D+a)}\cos bx: \quad D^2 \to -b^2 \text{ in } f(D+a)$$

GIVES The second half of the work when $V$ is trigonometric.

APPLIES After the shift; rationalise a leftover $D$ (class 004).

NOTE Use $b$ of $V$, not $a$ of the exponential: $e^x\cos x$: $D^2 \to -1$.

S006·6 · T006 00:04:20 · T006b 00:10:53 · CW U2 P17

### E1.4 · The Type 1 failure rule is a shift

ESA ★

$$\frac{e^{ax}}{(D-a)^k\,\psi(D)} = e^{ax}\,\frac{1}{D^k\,\psi(D+a)}\,1 = \frac{x^k}{k!}\,\frac{e^{ax}}{\psi(a)}$$

GIVES Why $a$ being a $k$-fold AE root brings in $x^k$.

APPLIES $V = 1$ and $f(a) = 0$ ($\psi(a) \neq 0$).

NOTE Same answer as $x\frac{e^{ax}}{f'(a)}$, $x^2\frac{e^{ax}}{f''(a)}$ of class 003.

ADDED FOR CLARITY S006·14

## 2 Type 4 with a polynomial: $x^m e^{ax}$ and mixed right-hand sides

### E2.1 · Then $V = x^m$ (Type 3)

ESA ★★

$$\frac{1}{f(D)}\left(x^m e^{ax}\right) = \frac{e^{ax}}{b_0}\left[1 \pm \varphi(D)\right]^{-1} x^m, \quad f(D+a) = b_0\left[1 \pm \varphi(D)\right]$$

GIVES The PI of a polynomial times an exponential.

APPLIES After the shift, $f(D+a)$ has a constant term $b_0 = f(a) \neq 0$.

NOTE Expand up to $D^m$ only (class 005 series). The PI is $e^{ax}$ times a polynomial of degree $m$.

S006·11 · S006·12 · T006b 00:12:47 · T006b 00:13:06 · T006 00:12:09 · CW U2 P18

### E2.2 · When the shift leaves only $D^k$

ESA ★★

$$\frac{1}{(D-a)^k}\left(e^{ax}V\right) = e^{ax}\frac{1}{D^k}V = e^{ax}\underbrace{\int\cdots\int}_{k} V\,dx\cdots dx$$

GIVES The PI by $k$ integrations (no constants).

APPLIES $f(D) = (D-a)^k$, e.g. $D^2 - 2D + 1 = (D-1)^2$ with $a = 1$.

NOTE Integrate by parts for $x\cos x$; "without writing the constant of integration, that is very important".

S006·8 · S006·9 · S006·14 · T006 00:06:40 · T006 00:07:16 · T006b 00:14:42

### E2.3 · Integration by parts (for $\frac{1}{D}$ of $x\cos x$, $x\sin x$)

ESA ★★

$$\int uv\,dx = u\int v\,dx - \int\left(\int v\,dx\right)u'\,dx$$

GIVES $\int x\cos x\,dx = x\sin x + \cos x$, $\int x\sin x\,dx = -x\cos x + \sin x$.

APPLIES $\frac{1}{D}$ of $x$ times a sine or cosine.

NOTE Slide 006·9 integrates $x\sin x$ instead of $x\cos x$ (gap list).

S006·9 · T006 00:07:34 · T006 00:09:11

### E2.4 · Superposition

ESA ★★

$$\frac{1}{f(D)}(X_1 + X_2) = \frac{1}{f(D)}X_1 + \frac{1}{f(D)}X_2 = \mathrm{PI}_1 + \mathrm{PI}_2$$

GIVES The PI of a mixed right-hand side.

APPLIES $X$ = a Type 4 term plus a Type 1, 2 or 3 term.

NOTE Each term chooses its own rule; see the class 005 pack for more mixed questions.

T006 00:10:40 · ESA Jan-24 s3 2c · CW U2 P18

### E2.5 · Reminder: $\sin^2 x$ and a factor $D$ ESA ★★

$$\sin^2 x = \frac{1-\cos 2x}{2}, \qquad \frac{1}{D\,\psi(D)}\,k = \frac{kx}{\psi(0)}$$

GIVES The constant part of $\sin^2 x$ when $D$ is a factor of $f(D)$.

APPLIES ESA Jan-24 s3 2c: $f(D) = D(D-1)^2$.

NOTE Taught in the class 004 and 005 packs.

ESA Jan-24 s3 2c · CW U2 P18

## 3 The Type 5 rule: $X = xV$

### E3.1 · Type 5: $X = xV$ ESA ★★

$$\frac{1}{f(D)}(xV) = \left[\frac{x}{f(D)} - \frac{f'(D)}{\big(f(D)\big)^2}\right]V = x\,\frac{1}{f(D)}V - \frac{f'(D)}{\big(f(D)\big)^2}V$$

GIVES The PI of $x$ times a function (usually $x\sin ax$, $x\cos ax$).

APPLIES $X = xV$; $f'(D) = \frac{d}{dD}f(D)$.

NOTE $\frac{x}{f(D)}V$ means: apply $\frac{1}{f(D)}$ to $V$, then multiply by $x$.

S006·4 · T006 00:01:28 · CW U2 P19 · CW U2 P20

### E3.2 · Where Type 5 comes from ESA ★

$$f(D)(xU) = x\,f(D)U + f'(D)U \;\Rightarrow\; \frac{1}{f(D)}(xV) = x\frac{1}{f(D)}V - \frac{1}{f(D)}f'(D)\frac{1}{f(D)}V$$

GIVES E3.1, with $U = \frac{1}{f(D)}V$; constant-coefficient operators commute, so the last term is $\frac{f'(D)}{(f(D))^2}V$.

APPLIES Leibniz: $D^n(xU) = xD^nU + nD^{n-1}U$.

NOTE Not on the slides.

ADDED FOR CLARITY S006·4

### E3.3 · Shortcut: $x\sin ax$ as an imaginary part ESA ★

$$\frac{1}{f(D)}\,x\sin ax = \operatorname{Im}\!\left[e^{iax}\frac{1}{f(D+ia)}\,x\right], \qquad \frac{1}{f(D)}\,x\cos ax = \operatorname{Re}\Big[\,\ldots\,\Big]$$

GIVES The same PI by Type 4 (with $a \to ia$) and Type 3 ($V = x$).

APPLIES $f(D)$ has real coefficients.

NOTE Only for checking or for fast work; the slide method is E3.1.

ADDED FOR CLARITY S006·4 · CW U2 P19 · CW U2 P20

# B Symbol glossary

| Symbol | Name | Meaning | Appears in |
|---|---|---|---|
| $D$ | Differential operator | $\frac{d}{dx}$; $\frac{1}{D}$ = integration without a constant | E1.1, E2.2 |
| $f(D)$ | Operator polynomial | Left-hand side operator of $f(D)y = X$ | E1.1, E3.1 |
| $f(D+a)$ | Shifted operator | $f(D)$ with every $D$ replaced by $D+a$ | E1.1, E1.2 |
| $f'(D)$ | f dash of D | Derivative of the polynomial $f(D)$ with respect to $D$ | E3.1, E3.2 |
| $a$ | Exponent coefficient | Coefficient of $x$ in $e^{ax}$; in Type 1 and Type 2 also the frequency of $\sin ax$ (clash) | E1.1 |
| $b$ | Frequency of $V$ | $V = \sin bx$ or $\cos bx$; $D^2 \to -b^2$ after the shift | E1.3 |
| $V$ | Remaining factor | $V(x)$: what is left after $e^{ax}$ (Type 4) or $x$ (Type 5) is split off | E1.1, E3.1 |
| $\varphi(D)$ | Phi of $D$ | $f(D+a) = f(a)[1 \pm \varphi(D)]$, only positive powers of $D$ | E2.1 |
| $b_0$ | Constant of $f(D+a)$ | Equals $f(a)$; taken out before the series | E2.1 |
| $\psi(D)$ | Psi of $D$ | $f(D) = (D-a)^k \psi(D)$ with $\psi(a) \neq 0$ | E1.4 |
| Im, Re | Imaginary, real part | $x \sin ax = \mathrm{Im}(xe^{iax})$, $x \cos ax = \mathrm{Re}(xe^{iax})$ | E3.3 |
| $\mathrm{PI}_1$, $\mathrm{PI}_2$ | Partial PIs | PIs of the separate terms of $X$ | E2.4 |

***Remark.*** Capital $X$ is the right-hand side; small $x$ the variable. $V = V(x)$ is the factor left after $e^{ax}$ (Type 4) or $x$ (Type 5) is split off; the slides use $V$ and $V(x)$ for the same thing. The letter $a$ is the coefficient of $x$ in the exponent ("$a$ is the coefficient of $x$ in the exponential function"), not a coefficient of $f(D)$. $f'(D)$ is the derivative of the polynomial $f(D)$ with respect to $D$ (e.g. $f(D) = D^2 - 1$, $f'(D) = 2D$). $m$ is both the AE unknown and the degree in $x^m$; the two never appear in the same line.